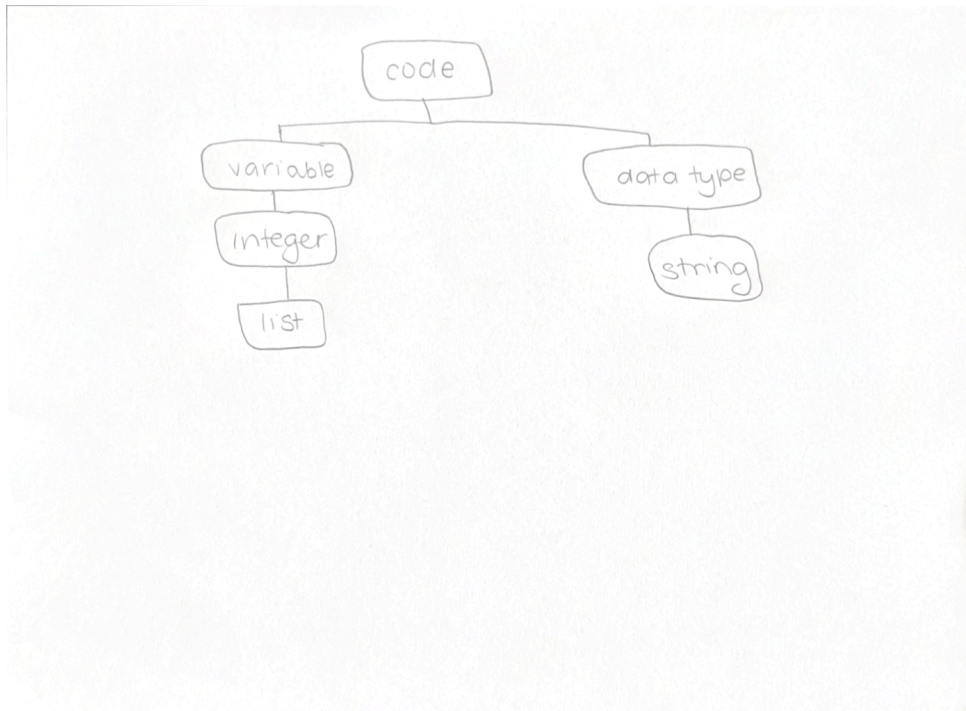
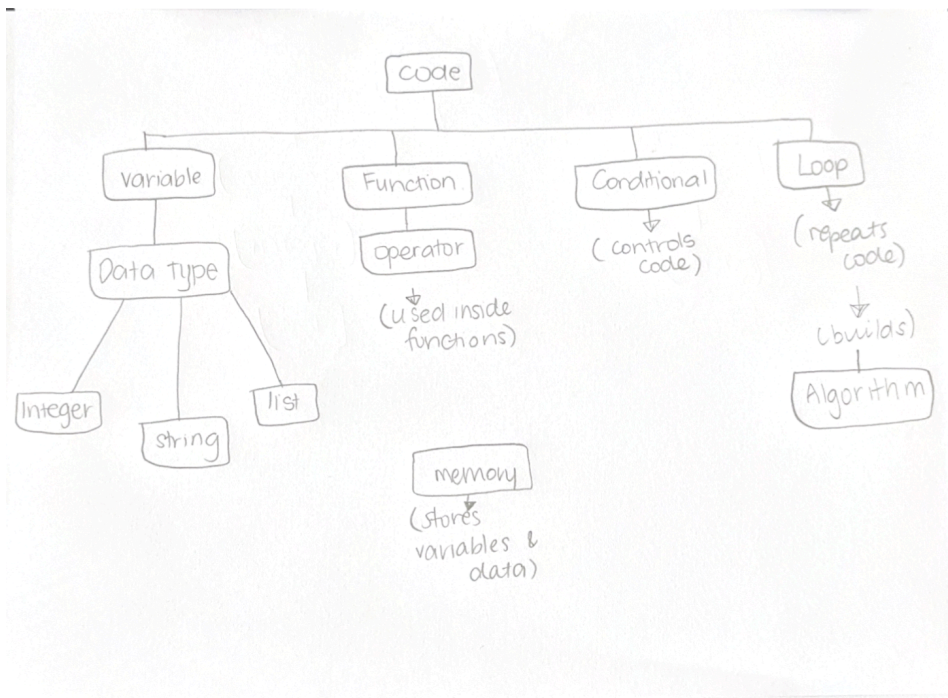


Appendix

Week 1



Week 12



A2 Project Journal — *Jazz Meets Code* (10-Week Log)

Purpose: Ongoing, honest record of progress while building a generative jazz-pop composition in TunePad. Captures feelings, ideas, code, screenshots, feedback, resources, and next steps. Contributes source material for the A2 Written Reflection.

How this journal is organised

- **Frequency:** Minimum once per week (often more). 10 weeks total.
 - **Artifacts to attach:** Code snippets, screenshots, audio exports, links, commit hashes, peer/tutor feedback, readings.
 - **Tags (suggested):** `#rhythm #swing #melody #chords #humanisation #bug #win #feedback #miro #todo`
 - **Miro board:** *(Paste link here)*
 - **GitHub repo:** *(Paste link here)*
-

Project snapshot (for quick context)

- **Project:** Generative jazz-pop composition (TunePad, Python)
 - **Core idea:** Blend jazz harmony + pop structure; add *human feel* via swing, micro-timing, and velocity variation.
 - **Target length:** ~2–3 minutes
 - **Success signals:** Feels *groovy* + *coherent*; clear code structure; reproducibility; reflection draws on concrete evidence.
-

Planned → Actual timeline at a glance

Week	Planned focus	Actual highlight	Evidence
1	Environment setup, scoping		
2	Chords & form sketch		
3	Bassline & groove prototype		
4	Drum swing & humanisation		
5	Melody generator v1		
6	Arrangement (A–B–Bridge)		
7	Refinement + FX		
8	User test + iterate		
9	Polish + documentation		
10	Final export + reflection notes		

Tip: Update the **Actual highlight** + **Evidence** cells each week with links, screenshots, and commit IDs.

Weekly entries

Structure for each entry

- *What I planned*
- *What I did (steps)*
- *Code snippet*
- *Screenshots / audio*
- *Feelings (surprises + frustrations)*
- *Peer/Tutor feedback*
- *What I learned (concepts)*

- *Next steps*
-

Week 1 — Setup, scoping, and rhythm references

What I planned

- Create TunePad project, repo, Miro board. Define scope, constraints, and “groove” reference tracks.

What I did (steps)

1. Created TunePad project (*paste link*).
2. Set BPM experiments (90–112) and settled on **96 BPM** for a relaxed pop-jazz feel.
3. Drafted pop form: **Intro–A–A–B–A–Outro**, 8-bar sections.
4. Sketched chord palette (key of F major / D minor options).
5. Wrote checklists for weekly increments.

Code snippet (tempo + form markers, pseudo-structure)

```
# tempo + section markers
bpm = 96 # set project tempo in TunePad UI or function
sections = {
  "intro": 0,
  "A1": 8,
  "A2": 16,
  "B": 24,
  "A3": 32,
  "outro": 40,
}
```

Screenshots / audio

[Insert: project home screenshot]

[Insert: section labels]

Feelings

Excited but slightly anxious about keeping code clean while experimenting musically.

Feedback

Peer suggested 96–100 BPM; tutor approved starting with clear sectioning.

What I learned

Planning sections early reduces later arrangement pain; commit small, often.

Next steps

Harmonise A-section with jazz-flavoured changes that still feel pop.

Week 2 — Chords & pop-friendly harmony

What I planned

Draft A-section chord loop; test cadence options and voice-leading.

What I did (steps)

1. Selected **Fmaj7 – Gm7 – Am7 – Gm7** for A-section (2 bars each).
2. For B-section, tried **Dm7 – G7 – Cmaj7 – C7** for tension-release into A.
3. Encoded chords as lists of MIDI pitches (root–3rd–5th–7th).

Code snippet (harmonic layer, illustrative)

chord tones (MIDI numbers for F major context)

Fmaj7 = [65, 69, 72, 76]

Gm7 = [67, 70, 74, 77]

Am7 = [69, 72, 76, 79]

Cmaj7 = [60, 64, 67, 71]

Dm7 = [62, 65, 69, 72]

G7 = [67, 71, 74, 77]

A_prog = [Fmaj7, Gm7, Am7, Gm7]

B_prog = [Dm7, G7, Cmaj7, Cmaj7]

(In TunePad, schedule these over bars with chosen instrument)

Screenshots / audio

[Insert: chord voicing piano roll]

[Audio: A vs B comparison]

Feelings

The B-section finally “lifts”; earlier tries were too bland.

Feedback

Peer: add 9ths occasionally for colour. Noted to sprinkle later.

What I learned

Voice-leading > fancy chords; keep pop clarity.

Next steps

Design bassline that locks the groove.

Week 3 — Bassline groove prototype

What I planned

Write bass patterns anchoring roots, adding approach notes into chord changes.

What I did (steps)

1. Root-focused eighth-note pattern with pickups into bar lines.
2. Chromatic approaches (e.g., $E \rightarrow F$) to add movement.
3. Tested range (E2–A2) to avoid muddiness.

Code snippet (bass, illustrative)

```
# bass pattern builder
import random

def walk_to(target, length=8):
    # simple approach: step towards target with small leaps
    cur = target - 5
    line = []
    for _ in range(length-1):
        step = random.choice([-2, -1, 1, 2])
        cur += step
        line.append(cur)
    line.append(target)
    return line

# schedule per chord root
roots = [65, 67, 69, 67] # F, G, A, G (MIDI)
A_bass = [walk_to(r) for r in roots]
```

Screenshots / audio

[Insert: bass lane screenshot]

Feelings

Hearing bass + chords together = first *“it’s a song”* moment.

Feedback

Tutor: ensure bass doesn't clash with kick accents.

What I learned

Chromaticism adds life; keep pattern density controlled.

Next steps

Add drums with swing and velocity humanisation.

Week 4 — Drums, swing, and micro-timing

What I planned

Program a pop-leaning kit with jazz ride feel; experiment with swing %.

What I did (steps)

1. Kick on 1 & 3, backbeat snare on 2 & 4; light ride pattern (jazz eighths).
2. Implemented swing by delaying off-beats; added small $\pm$ timing jitter.
3. Varied hi-hat/ride velocities to breathe.

Code snippet (swing & humanisation, illustrative)

```
import random
```

```
SWING = 0.58 # 0.5 = straight, >0.5 = swung
```

```
JITTER_MS = 8
```

```
# pseudo: schedule_note(inst, pitch, bar, beat, dur, vel, offset_ms)
```

```
# apply swing: if beat is an off-eighth, push later by (SWING-0.5)
```

Screenshots / audio

[Insert: drum grid + swing settings]

Feelings

Swing at 60% felt too lumpy; 58% sits nicer with pop backbeat.

Feedback

Peer: lighten snare velocity; ride pattern is lovely.

What I learned

Micro-timing works best when subtle; velocity shaping matters.

Next steps

Build a simple melody generator with motifs.

Week 5 — Melody generator v1 (motifs)

What I planned

Create a motif-based generator that respects chord tones + passing notes.

What I did (steps)

1. Defined two short motifs (up a 3rd; enclosure into target).
2. Chose note pools: chord tones + scale tones; occasional chromatic approach.
3. Added “rest probability” to avoid over-playing.

Code snippet (motif engine, illustrative)

```
import random

def motif_up_third(root):
    return [root, root+4, root+7] # simple pop-friendly triad arc

def motif_enclose(target):
    return [target-1, target+1, target]

REST_P = 0.15

# build 2-bar phrases choosing motifs conditioned on chord
```

Screenshots / audio

[Insert: melody lane]

Feelings

First melodies sounded “too busy”; rests fixed it.

Feedback

Tutor: Keep melodic range singable; think call-and-response with bass.

What I learned

Less is more. Silence is part of groove.

Next steps

Arrange sections A–A–B–A with fills leading into transitions.

Week 6 — Arrangement (A–A–B–A + fills)

What I planned

Structure full form; add 1–2 bar fills into B and final A.

What I did (steps)

1. Muted melody in A1 to introduce gradually.
2. Added drum fill (snare/hat 16ths) into B; bass pickup into final A.
3. Subtle pad on B for lift; removed in outro for intimacy.

Code snippet (arranger, illustrative)

enable/disable tracks per section

arrangement = {

```
"intro": {"chords": True, "bass": False, "drums": True, "melody": False},
"A1":    {"chords": True, "bass": True, "drums": True, "melody": False},
"A2":    {"chords": True, "bass": True, "drums": True, "melody": True },
"B":     {"chords": True, "bass": True, "drums": True, "melody": True },
"A3":    {"chords": True, "bass": True, "drums": True, "melody": True },
"outro": {"chords": True, "bass": False, "drums": True, "melody": False},
}
```

Screenshots / audio

[Insert: arrangement view + automation]

Feelings

The form breathes; B-section feels like a chorus now.

Feedback

Peer: Maybe high-pass pad to avoid muddying bass.

What I learned

Sparse intros make the first drop more satisfying.

Next steps

Refine FX + humanisation; add tasteful reverb/width.

Week 7 — Refinement, FX, and human feel pass

What I planned

Tweak velocities, reverb sends, and micro-timing; add occasional chord extensions.

What I did (steps)

1. Added 9ths to Fmaj7 and Cmaj7 on every second loop.
2. Ride pattern alternates bar-to-bar; ghost notes on snare.
3. Tightened bass timing on downbeats; loosened off-beats slightly.

Code snippet (variations, illustrative)

```
# dynamic chord colouring  
Fmaj9 = [65, 69, 72, 76, 79]  
Cmaj9 = [60, 64, 67, 71, 74]
```

```
# choose extension every second loop
```

Screenshots / audio

[Insert: FX panel / send levels]

Feelings

Small changes = big feel upgrade.

Feedback

Tutor: Great restraint; keep tails short to preserve groove.

What I learned

Human feel emerges from *consistent* small deviations.

Next steps

Prepare for user test; create short survey prompts.

Week 8 — User test (mini) + iteration

What I planned

Conduct 5-minute listens with 2–3 peers; collect “groove feel” ratings.

What I did (steps)

1. Played A–B–A; asked: “Does this feel human or rigid?” (1–5 scale).
2. Logged comments on melody density and drum brightness.

3. Adjusted ride EQ and reduced melody notes in A2.

Code snippet (NA)

Focus this week: listening and tweaks.

Screenshots / audio

[Insert: survey screenshot or notes]

Feelings

Constructive feedback; surprised how much brightness affects perceived swing.

Feedback (summarised)

- “Ride a touch harsh at times.”
- “Melody lands nicer when it answers bass.”

What I learned

Perceived groove is tonal as well as rhythmic.

Next steps

Polish code for readability + comments; prep exports and repo.

Week 9 — Polish, documentation, exports

What I planned

Clean code; add docstrings and section headers; export audio.

What I did (steps)

1. Consolidated functions and named constants.
2. Added **README** run-instructions + dependency notes.
3. Exported WAV/MP3 snippets (A, B, full).

Code snippet (module header, illustrative)

```
"""
```

```
Jazz Meets Code — Generative jazz-pop composition
```

```
- Requires: TunePad environment
```

```
- How to run: open project → run cells top-to-bottom → press Play
```

```
"""
```

BPM = 96
KEY = "F major"
SEED = 42 # for reproducibility

Screenshots / audio

[Insert: code with comments]
[Insert: export settings]

Feelings

Happy with clarity; code reads like a story.

Feedback

Peer: README is clear; add screenshot of the UI.

What I learned

Documentation is part of design.

Next steps

Final QA listen + reflection notes.

Week 10 — Final QA + reflection capture

What I planned

QA session on headphones and speakers; capture reflection snippets while fresh.

What I did (steps)

1. Balanced levels; confirmed no clipping; checked section lengths.
2. Wrote bullet reflections on learning (see below).
3. Prepared demo for 3-minute explainer video.

Code snippet (NA)

Music + documentation focus this week.

Screenshots / audio

[Insert: final waveform]

Feelings

Proud of the groove; feels musical, not mechanical.

Feedback

Tutor sign-off on readiness.

What I learned (headlines)

- Pop clarity + jazz colour is a balancing act.
- Micro-timing & velocity shape *feel* more than I expected.
- Planning sections early sped up arranging.

Next steps

Submit prototype; write full A2 reflection using this journal.

Evidence index (fill as you go)

- **Commits:** paste short hashes + notes each week
 - **Screenshots:** paste thumbnails or links
 - **Audio exports:** paste file links
 - **Peer feedback docs:** paste Sheets/Forms links
 - **Readings/resources:** paste links + 1-line relevance
-

Reflection seed notes (for A2 Written Reflection)

- **Data/code literacy:** Explain how motif functions, humanisation params, and arrangement toggles demonstrate code control and musical intention.
 - **Surprises:** Tonal brightness changed groove perception as much as timing.
 - **Frustrations:** Over-busy melodies; initial swing set too high.
 - **Decisions:** Kept BPM 96; limited melody density; added 9ths sparingly.
 - **Future work:** Parameter UI for live swing/velocity tuning; more motif types.
-

Appendices

A. Running notes template (copy weekly)

- Todo:
- Blockers:
- Quick wins:
- Risk/mitigation:

B. Feedback log template

- Date / From / Summary / Action taken

C. Checklist before submission

- Code runs top-to-bottom without errors
- Comments + docstrings present
- Audio export attached
- Journal updated with screenshots + links
- Repo README complete

Reminder: Keep this living. Add real screenshots, links, and commit IDs. Your future reflection will thank you!